

Quantifying and Eliminating Sources of Uncertainty in Ecological Dynamics Models

Vivian Li, Luis de Pablo, Henry Li, Laura Dee, Aaron Clauset
Department of Computer Science, University of Colorado Boulder

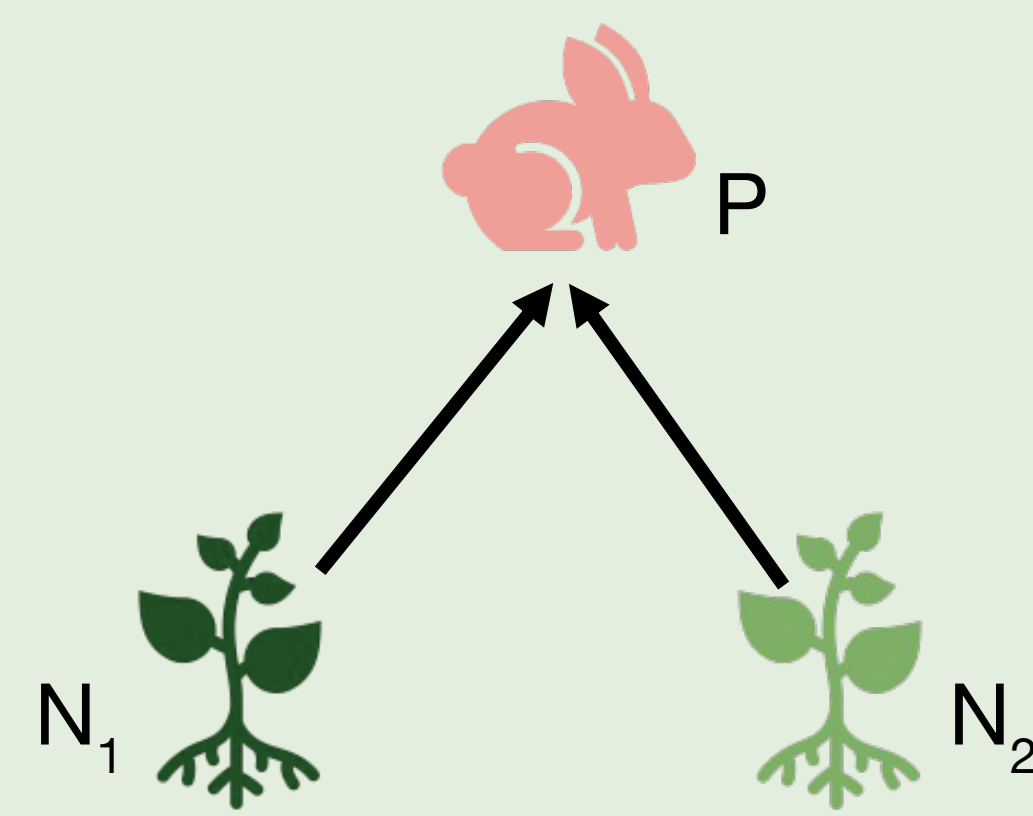
Background

- Ecological systems can be modeled as dynamical systems in which species abundances evolve according to coupled differential equations defined by an interaction network.
- These models are used to study ecosystem stability¹, predict response to future disturbances², and guide possible management interventions³.
- Model complexity varies with spatial and temporal scale. Increasing model complexity requires assumptions about interaction structure, functional forms, and parameter values.
- Due to poor quality of available data, ecological predictions are often limited by uncertainty in both model structure and parameterization.

Creating an Ecosystem Model

Interaction Network → Adjacency Matrix

Species interaction networks are constructed from a combination of empirical observation, prior studies, and indirect inference. While interaction presence can sometimes be identified, interaction strengths are rarely well constrained and are often assumed equal, estimated from limited experiments, or randomly assigned in modeling studies.



	P	N ₁	N ₂
P	0	0	0
N ₁	1	0	0
N ₂	1	0	0

Adjacency Matrix → Dynamical System

Ecological dynamics are modeled using systems of equations derived from the interaction matrix. General frameworks such as Lotka-Volterra models provide a common starting point, but are frequently modified to emphasize system-specific processes (e.g. fisheries, microbial communities, plant interactions).

$$\Delta \text{predator biomass } \left(\frac{dP}{dt} \right) = \underbrace{\beta c P N_1}_{\text{consumption}} + \underbrace{\beta c P N_2}_{\text{consumption}} - \underbrace{m P}_{\text{mortality}}$$

$$\Delta \text{prey biomass } \left(\frac{dN_i}{dt} \right) = \underbrace{r_i N_i}_{\text{intrinsic growth}} \left(1 - \frac{N_i}{K_i} \right) - \underbrace{\beta N_i P}_{\text{predation}}$$

Small uncertainties during modeling can lead to qualitatively different ecological outcomes, making prediction inseparable from uncertainty quantification.

References

- ¹Domínguez-García *et al.* (2019) *PNAS* 116.
²Bode *et al.* (2017) *Methods in Ecology and Evolution* 8.
³Peterson *et al.* (2021) *Journal of Applied Ecology* 58.

Methods

- We constructed a minimal three-species dynamical system consisting of one predator (P) and two prey (N_1, N_2) using generalized Lotka-Volterra equations.
- To isolate the effects of uncertainty, predator-prey interactions were held fixed, while interactions between the two prey species were systematically varied.
- Structural uncertainty in the interaction network** was introduced by allowing prey-prey interactions to be positive, negative, or absent, corresponding to facilitation, competition, or no interaction.
- Parametric uncertainty** was explored by sampling interaction strengths across a range of magnitudes for each allowed interaction type.
- Model uncertainty** was introduced by altering how prey-prey interactions entered the dynamical equations. Specifically, prey-prey interactions were implemented through four alternative mechanisms:

1. Modification of intrinsic growth rate

$$\frac{dN_i}{dt} = (r_i + \alpha_{ij} B_j) B_i \left(1 - \frac{B_i}{K_i} \right) - \beta B_i B_p$$

2. Modification of carrying capacity

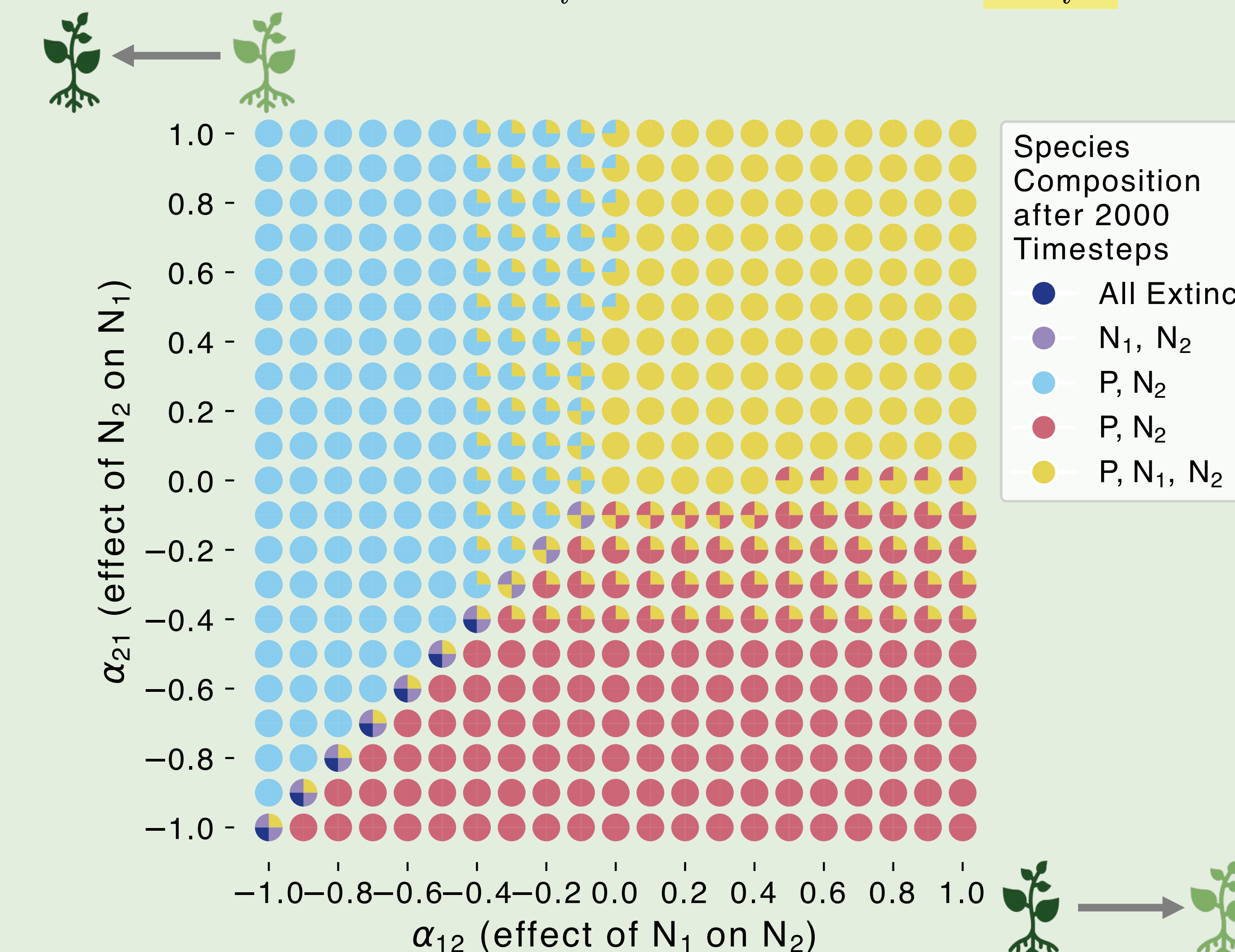
$$\frac{dN_i}{dt} = r_i B_i \left(1 - \frac{B_i - \alpha_{ij} B_j K_i}{K_i} \right) - \beta B_i B_p$$

3. Modification of total biomass

$$\frac{dN_i}{dt} = r_i B_i \left(1 - \frac{B_i}{K_i} \right) - \beta B_i B_p + \frac{\alpha_{ij} B_j}{B_i}$$

4. Combined effects on carrying capacity and biomass

$$\frac{dN_i}{dt} = r_i B_i \left(1 - \frac{B_i - \alpha_{ij} B_j K_i}{K_i} \right) - \beta B_i B_p + \frac{\alpha_{ij} B_j}{B_i}$$



Simulations were run across a range of prey-prey interaction values and model structures until equilibrium. Final species composition is shown across parameter space, with each point partitioned into four quadrants representing the outcomes of Eqs. 1-4 (clockwise from top right).

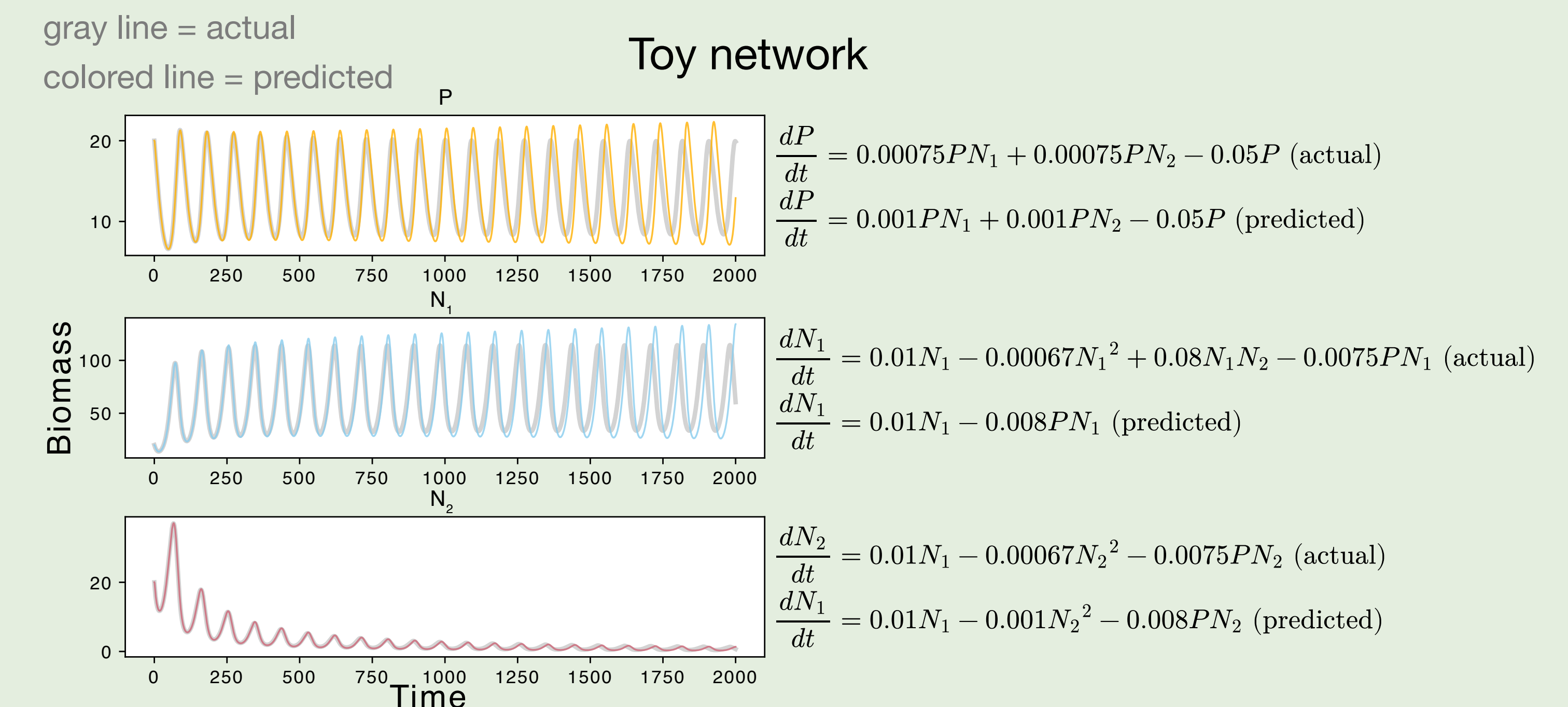
Eliminating Uncertainty

Sparse Identification of Nonlinear Dynamics (SINDy)

Selects from a library of nonlinear candidate functions (e.g. polynomials) to reverse-engineer system dynamics.

Input: time-series abundance data

Output: differential equations describing system dynamics

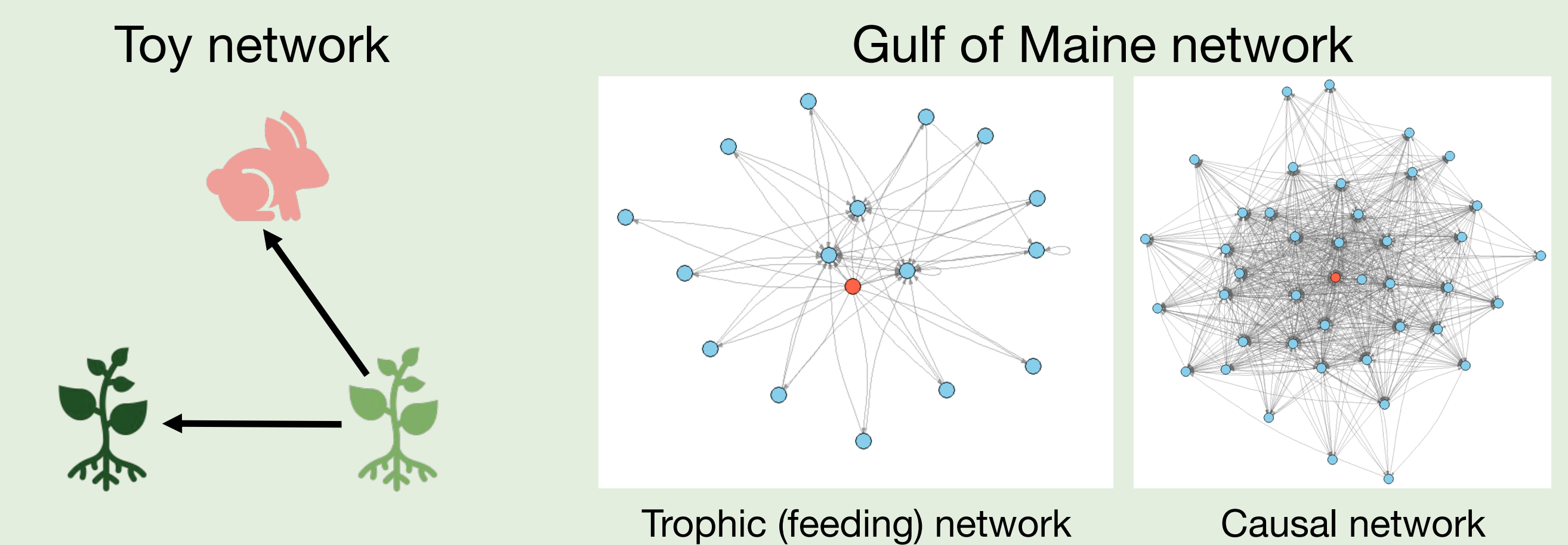


Convergent Cross Mapping (CCM)

Tests pair-wise interactions to build a causal network between nodes.

Input: time-series abundance data

Output: a network representing unidirectional causality between each driver and response variable



Conclusion

We aim to understand how uncertainty propagates from ecosystems to interaction networks and ultimately to dynamical models and predictions. While methods such as SINDy and CCM can recover equations or network structures in small systems, these methods currently fail to scale to real ecosystems, where even dense time series and well-characterized food webs do not reproduce observed dynamics. Moving forward, we seek to develop scalable methods for inferring ecological interactions and propagating uncertainty through dynamical models, with the goal of determining not only how to improve ecological predictions, but also when model-based inference is reliable for guiding decision-making.